

The University of Hong Kong
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 DATA MINING

Date: 19th December 2019

Time: 6:30pm – 8:30pm

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidate's responsibility to ensure that his/her calculator operates satisfactorily, and candidate must record the name and type of the calculator used at the beginning of their answer script.

IMPORTANT NOTE: Please be reminded that you have agreed to observe academic integrity. In particular, your submission with this exam is solely your own and you have not received any unauthorized assistance in this exam including unauthorized input from the Internet.

Answer ALL 6 questions.

1. (a) (7%) Given a dataset D of records such that each record is associated with either the label "Class C_1 " or "Class C_2 ". With Naive Bayesian Classification, given a record X , explain what values are computed and how the classifier classifies X into one of the two classes based on the computed values.
(b) (7%) Explain how you would modify your Naive Bayesian Classifier if there are three classes C_1 , C_2 , and C_3 , instead of two.
2. (a) (6%) Explain the following methods of obtaining a training set and a test set for classifier evaluation: (1) Holdout, (2) Random Subsampling, (3) Cross Validation, (4) Bootstrap.
(b) (7%) If you have very limited amount of labeled data, which method will you choose and why?
3. (15%) Consider the set of transactions shown in Table 1. Assume a support threshold of 3 transactions, draw the FP-tree of the dataset. (Note: in FP-tree construction, frequent items are sorted in descending order of their frequencies. We break tie by alphabetical order of the frequent items. For example, if items a and b have the same frequency counts, then item a is ahead of b in the sorted order because a precedes b in alphabetical order.) Show how you use the FP-tree to deduce all the frequent patterns that involve item p .

Transaction ID	Items Bought
T_1	$\{f, a, c, d, g, i, m, p\}$
T_2	$\{a, b, c, f, l, m, o\}$
T_3	$\{b, f, h, j, o\}$
T_4	$\{b, c, k, s, p\}$
T_5	$\{a, f, c, e, l, p, m, n\}$
T_6	$\{d, c, s, h, j\}$
T_7	$\{d, o, f, c\}$

Table 1: Dataset 1

4. Consider the dataset shown in Table 2. In the dataset, there are 6 objects ($X_1, \dots, X_6$), each with 2 attributes (a_1, a_2) and a class label ($Class$).

Object	a_1	a_2	Class
X_1	20	20	C_1
X_2	27	21	C_2
X_3	21	24	C_1
X_4	36	21	C_2
X_5	35	21	C_3
X_6	22	20	C_1

Table 2: Dataset 2

- (a) (7%) If Attribute a_1 is used to derive the root test of a decision tree, determine the best split point. Assume that Attribute a_1 is numerical and $GINI$ is used to evaluate tests. Show your calculation.
- (b) (4%) Consider building another decision tree that uses Attribute a_2 as the root test. If we treat Attribute a_2 as an ordinal attribute, calculate the information gain of the test shown in Figure 1.

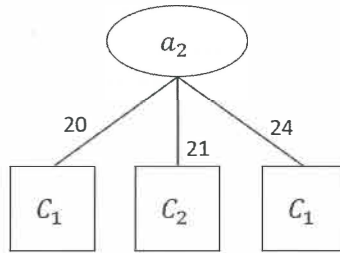


Figure 1: Classification tree

- (c) (6%) Use the test data set shown in Table 3 to evaluate the decision tree shown in Figure 1. Construct the confusion matrix, and then calculate the precision and recall of the target class C_1 .

Object	a_1	a_2	Class
X_7	22	21	C_1
X_8	21	20	C_1
X_9	23	24	C_3
X_{10}	24	24	C_2

Table 3: Test set

5. Consider the market basket dataset shown in Table 4.

Transaction ID	Items Bought
T_1	$\{a, b, c, e\}$
T_2	$\{a, d, e, f\}$
T_3	$\{b, c, d, f\}$
T_4	$\{b, c, e, f\}$
T_5	$\{b, c, f\}$

Table 4: Dataset 3

- (6%) Given transaction $X = \{a, f\}$, calculate the Jaccard coefficients between X and each of T_1 , T_2 , and T_3 . Rank transactions T_1 to T_3 according to their similarities to transaction X . List the most similar transaction first.
 - (6%) Given support threshold $\rho_s = 30\%$ and confidence threshold $\rho_c = 70\%$. List all frequent 3-itemsets.
 - (4%) Consider rules of the form $X \rightarrow Y'$, which is interpreted as “If a customer buys all items in X , then he/she is unlikely to buy any items in Y .” Illustrate how the dataset shown in Table 4 should be transformed to find those rules.
 - (6%) Calculate the confidence and the lift of the rule $\{f\} \rightarrow \{d\}'$. Will you consider this a good rule? Why?
6. Table 5 shows the proximity matrix of 4 objects.

	O_1	O_2	O_3	O_4
O_1	0	0.6	0.1	0.9
O_2	0.6	0	0.3	0.4
O_3	0.1	0.3	0	0.2
O_4	0.9	0.4	0.2	0

Table 5: Proximity matrix of 4 objects

- (4%) Find the medoid of the dataset.
- (7%) Cluster the objects using Complete-Link clustering. Show your results by drawing a dendrogram. Mark the dendrogram with the proximity values of the merge points.
- (8%) Compare k -means with hierarchical agglomerative clustering. Discuss the pros and cons of the two approaches.

END OF PAPER